

Linus Bierhoff

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WORK EXPERIENCE

h-lab @ Karlsruhe Institute of Technology, Karlsruhe, Germany <i>Junior Researcher – Lead Engineer & Information Systems Research</i>	September 2025 – Present
- Lead the technical roadmap for AMSL (https://amsl.app), an EdTech platform serving 1,200+ students over 3 years, prioritizing feature execution based on user feedback and learning analytics. - Architected a domain-specific YAML-based Low-Code framework, enabling non-technical educators to deploy custom educational RAG agents; successfully adopted by 200+ students within the first year - Co-authored a paper accepted at CHI 2026 (Top-tier HCI)	
Sustainability Innovation @ SAP SE, Karlsruhe, Germany <i>Internship – Applied AI</i>	March 2025 - August 2025
- Secured Finalist status (Top 5%) out of 100+ developer teams in the "Agentic AI Days" hackathon by engineering an autonomous agent for sustainable product design - Lead the development for an advanced agentic pipeline initiative (LangGraph) within SAP Sustainability; defined the technical roadmap and personally executed the core implementation, translating high-level research goals into a functional prototype that enhances information retrieval efficiency Pitched the autonomous agent to the Global VP of Sustainability Innovation; this secured the project's place in the Chief Product Officer's SAP Sapphire 2025 roadmap and enabled me to showcase the solution directly to customers at the SAP-KIT Research and Innovation Symposium 2025 Conducted a comparative analysis of multi-agent implementation patterns and presented findings to key stakeholders, directly influencing the architectural choices of 3 internal production teams	
h-lab @ Karlsruhe Institute of Technology, Karlsruhe, Germany <i>Student Assistant – Full-Stack Software Development</i>	February 2023 - February 2025
Engineered a RESTful API using Rust for the AMSL Platform (https://amsl.app) enabling real-time chat features to be delivered reliably to a user base of 1,200+ students with a burst-traffic during live lectures up to 200 participants Built a responsive cross-platform mobile frontend using Dart & Flutter, enabling a seamless native experience across both iOS and Android devices Orchestrated containerized deployment and CI/CD via Kubernetes, ensuring system reliability and continuous delivery for the student user base	

EDUCATION

KIT - Karlsruhe Institute of Technology, Karlsruhe, Germany <i>M.Sc. in Information Systems</i>	October 2025 – Present
- Focus on Natural Language Processing, Artificial Intelligence, Interactive Systems and Statistics - Current Grade: 1.3 (German Scale 1-5)	
KIT - Karlsruhe Institute of Technology, Karlsruhe, Germany <i>B.Sc. in Information Systems</i>	October 2021 – March 2025
- Focus on: Artificial Intelligence, Human Computer Interaction, Statistics and Management & Finance - Thesis: <i>Building and evaluating an LLM-based tutor for enhancing post-lecture studying</i> - Grade: 1.5 (German Scale 1-5) (Top 10%)	
Abitur / A-level equivalent. <i>Grade: 1.4 (German Scale 1-5)</i>	August 2013 - June 2021

ACADEMIC & PERSONAL PROJECTS

Advanced Agentic Pipelines & Prompt Optimization - Python, LangGraph, RAG, MCP, DSPy, MLflow	
- Designed and deployed multiple autonomous agent pipelines, utilizing frameworks like LangGraph and building tools with MCP, for video generation, information extraction and database manipulation - Utilizing DSPy and MLflow for programmatic prompt optimization to systematically improve model output	
Open Data Autonomous Analyst - Python, LangChain, RAG, SQL	
- Awarded 2nd Place @ Data Excellence Talent Award 2024 (Team Lead) - Engineered a hybrid Agentic RAG system using LangChain to bridge semantic metadata retrieval with SQL, enabling autonomous fetching and hydration of 1.6M Open Data datasets for advanced analysis	
Autonomous Drone Navigation System - Python, PyTorch, OpenCV	
- Developed an autonomous flight control using OpenCV and ArUco markers Trained Convolutional Neural Networks (CNNs) using PyTorch and successfully deploying models for real-time obstacle detection	

SKILLS

Software Engineering	Rust, Python, Dart, Java, SQL, TypeScript
ML & Agentic Frameworks	LangGraph, Haystack, DSPy, MLflow, PyTorch, Scikit-learn, Pandas, NumPy
Infrastructure & DevOps	Git, CI/CD, Docker, Kubernetes
Methodologies	Scrum (Foundations), Design Thinking, Kanban
Spoken Languages	German (native), English (business proficiency)
Interests	Tennis (Club player), Running (Half-Marathon focus), Specialty Coffee